

Universal statistics of hippocampal place fields across species and dimensionalities

Highlights

- We present a model that explains irregular CA1 responses in large environments
- The theory explains statistics of field sizes, arrangement, and shape heterogeneities
- Predictions are quantitative, in 1/2/3D spaces of varying size, in rodents and bats
- CA1 inputs are modeled as Gaussian processes, indicative of random connectivity

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In brief

Mainali et al. show that a simple mathematical model explains recent observations of heterogeneous and irregular spatial responses in the hippocampus in large environments. The model unifies observations from environments of varying sizes and dimensions in multiple species, suggesting that a common mechanism underlies the hippocampal spatial code.



Article

Universal statistics of hippocampal place fields across species and dimensionalities

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SUMMARY

Hippocampal place cells have single, bell-shaped place fields in small environments. Recent experiments, however, reveal that, in large environments, place cells have multiple fields with heterogeneous shapes and sizes. We show that this diversity is explained by a surprisingly simple mathematical model, in which place fields are generated by thresholding a realization of a random Gaussian process. The model captures the statistics of field arrangements and generates new quantitative predictions about the statistics of field shapes and topologies. These predictions are quantitatively verified in bats and rodents, in one, two, and three dimensions, in both small and large environments. These results imply that common mechanisms underlie the diverse statistics observed in different experiments and further suggest that synaptic projections to CA1 are predominantly random.

INTRODUCTION

A central goal of systems neuroscience is to characterize how neural systems represent information about the world or about the brain's internal state. For several decades, much of the thinking about neural population codes was motivated by reports on neurons with highly stereotyped tuning functions. Neurons were often observed to have a smooth, typically unimodal tuning to the encoded variable, centered on preferred stimuli that vary across the neural population.^{1,2} Examples include the radially symmetric receptive fields of retinal ganglion cells,³ the receptive fields of simple cells in the primary visual cortex,^{4,5} cosine tuning to movement direction in the motor cortex,⁶ the unimodal fields of head direction cells,⁷ and the fields of hippocampal place cells in small environments.⁸

Experiments in the past decade, however, have uncovered neural response functions that are much less stereotyped and regular than observed previously.^{9–17} These recent findings were enabled by high-throughput recording techniques, which substantially reduced bias in the selection of cells for analysis, as well as new technologies that enabled the monitoring of neural activity in freely behaving animals, under reduced behavioral constraints.^{18–21}

Some of the most striking examples of irregular neural responses were recently identified in area CA1 of the hippocam-

pus. The classical view of spatial coding by place cells in this area has been that they are active in a single, compact region of space and exhibit a stereotyped bell-shaped tuning to position.^{8,22} Several recent experiments in bats and rodents have revealed, however, that this picture breaks down in large environments.^{9,10,21,23–25} Place cells typically fire in multiple locations, and furthermore, the multiple firing fields of individual cells, as well as those of the whole population, vary in size and shape, the latter of which can deviate substantially from the classical bell-shaped form. It remains unknown, however, whether the irregular statistics of place fields in bats and rodents can be described in terms of common mathematical principles and whether these statistics yield insights on the synaptic architecture that underlies spatial coding in hippocampal area CA1.

A number of theoretical studies have modeled the spatial selectivity of place cells in large environments. In one approach, heterogeneous activity patterns were embedded in the attractor manifolds of recurrent neural networks, serving as models of place-cell dynamics in area CA3. Place cells were modeled as having multiple, randomly positioned firing fields^{26,27} with variable peak firing rates and field sizes.²⁷ Another study²⁸ predicted an exponential distribution of place-field sizes based on a normative theory. Finally, multi-peaked, multi-scale tuning curves were shown to be advantageous for coding efficiency.^{21,29} However, there is a lack of a theory that can explain



the observed heterogeneity of place-field properties, across the diverse experimental conditions, from basic principles.

Here, we report that a surprisingly simple generative model accounts for highly detailed features of place-field statistics. We model place fields as derived from a realization of a random Gaussian process over the spatial coordinates: a realization of the process is sampled for each cell and is then thresholded and rectified, resulting in multiple, heterogeneous fields (Figure 1A). A Gaussian process is a random function whose values at any discrete set of positions are jointly normal.³⁰ The statistics of a Gaussian process are uniquely determined by its mean, which we set to zero without loss of generality as its choice is redundant with the choice of the threshold, and by the spatial correlation function, which we assume to be translationally invariant.

Several lines of reasoning have led us to examine the above model. First, a Gaussian process is the random process that maximizes entropy under constraints on the mean and the spatial covariance function. Hence, our modeling choice invokes minimal assumptions other than a specification of a place cell's underlying input correlation structure.³¹ Second, Gaussian statistics arise when many independent random variables are summed. The summed presynaptic input into area CA1 from area CA3 and entorhinal cortex is thus expected to be approximately Gaussian if synaptic weights are predominantly random (Figure 1; Methods S1 and S2). Third, a neural code structured as a random Gaussian process can achieve exponential coding efficiency.²⁹ We further discuss these motivations in the discussion.

RESULTS

Field arrangements

Place-field locations in the model are identified as the regions in space in which the Gaussian process exceeds a given threshold (Methods S1). Extensive results exist for the statistics of these regions in the mathematical literature, where they are called "excursion sets."³⁰ A recent result of central importance to this work is that threshold-crossing statistics assume a universal form when the spatial correlation function of the process obeys mild smoothness requirements and the threshold is sufficiently high.³² Threshold-crossing statistics are then insensitive to the detailed structure of the correlation function and depend only on two scalar parameters: the normalized threshold, θ , defined as the ratio between the threshold and the standard deviation of the Gaussian process (Equation 3) and a correlation length, σ , which is derived from the spatial correlation function (Equation 4).

Specifically, the mean density of fields and the mean field size can both be expressed in terms of σ and θ using the Kac-Rice formula³³ for the threshold crossings of a Gaussian process (Methods S3, see Equations S12 and S14). Furthermore, the full distribution of field sizes acquires a universal form in the limit of large θ (Methods S4), expressed as

$$P(s) = \frac{2\beta}{D} s^{\left(\frac{D}{2}-1\right)} \exp(-\beta s^{\frac{D}{2}}), \quad (\text{Equation 1})$$

where D is the dimensionality of the space over which the Gaussian process is defined, s is the field size (length in one

dimension, area in two dimensions, and volume in three dimensions; 1D, 2D, and 3D hereafter), and the single parameter β can be expressed in terms of σ and θ (see Methods S3, Equation S18). In the 1D case ($D = 1$), this expression takes the form of a Rayleigh distribution (Figure 1C). Similarly, field gap statistics assume a universal form in the high-threshold limit: threshold crossings are spatially uncorrelated and, consequently, gap sizes follow an exponential distribution with a mean that can be expressed in terms of σ and θ (see STAR Methods). Thus, according to the model, the empirical distributions of CA1 field sizes and gaps between fields should be jointly explained using the two parameters, σ and θ .

We began by examining place fields recently recorded in bats flying in a 200-m-long tunnel, where the highly stereotyped trajectory of the animals along the 1D track allowed for the gathering of comprehensive statistics on the spatial arrangement of place fields.²¹ We determined the two parameters σ and θ by matching the means of the field sizes and gaps to the model in the data from the 200-m-long tunnel. This fitting procedure, based only on the means, was sufficient to capture the full distributions of field sizes and gaps (Figures 1C and 1D). This outcome was particularly striking for the field sizes, whose distribution had a distinctive, highly asymmetric form.

Due to its highly asymmetric structure (Figure 1C), the field-size distribution was previously fitted heuristically to a log-normal distribution, yet this approach lacked a principled rationale. By contrast, in our model, the asymmetry is explained by the statistical dependencies between locations of adjacent threshold crossings of the underlying random Gaussian process³⁴ (see Methods S4). The empirical field-size distribution was more likely to arise from the Gaussian threshold-crossing model than the best-fit log-normal distribution, even though the model had one fewer degree of freedom (assessed via likelihood and parameter-aware likelihood based model performance metrics: $\Delta\text{LLR} < 0$, $\Delta\text{BIC} < 0$, $\Delta\text{AIC} < 0$; see STAR Methods). Furthermore, skew and kurtosis of the log-field-size distribution were in agreement with the threshold-crossing model, but not with the log-normal distribution, according to which they should be close to zero (Figure 1E; see STAR Methods). Thus, the model captured subtle features of the distribution beyond its first- and second-order moments. As expected in the high-threshold limit, the distribution of gap sizes followed an exponential distribution (Figure 1D).

We subsequently tested the model's ability to explain the statistics of place fields measured in rodents running in long 1D tracks. Field size distributions extracted from experiments in which mice ran in a 40-m-long 1D virtual track²⁴ were in qualitative agreement with the Rayleigh distribution (Figure 1F). We also analyzed the distribution of field sizes from an experiment in which rats explored a 48-m-long 1D maze.¹⁰ The distribution of field sizes was in quantitative agreement with the distribution predicted by the model, with matching skew and kurtosis (Figure S1).

Place fields in two and three dimensions

Experimental data on place fields in rats navigating large 2D environments²⁵ and bats in 3D rooms²³ reveal place cells with multiple, heterogeneous place fields. Intriguingly, though, the experiments uncover field-size distributions that are distinct in one,

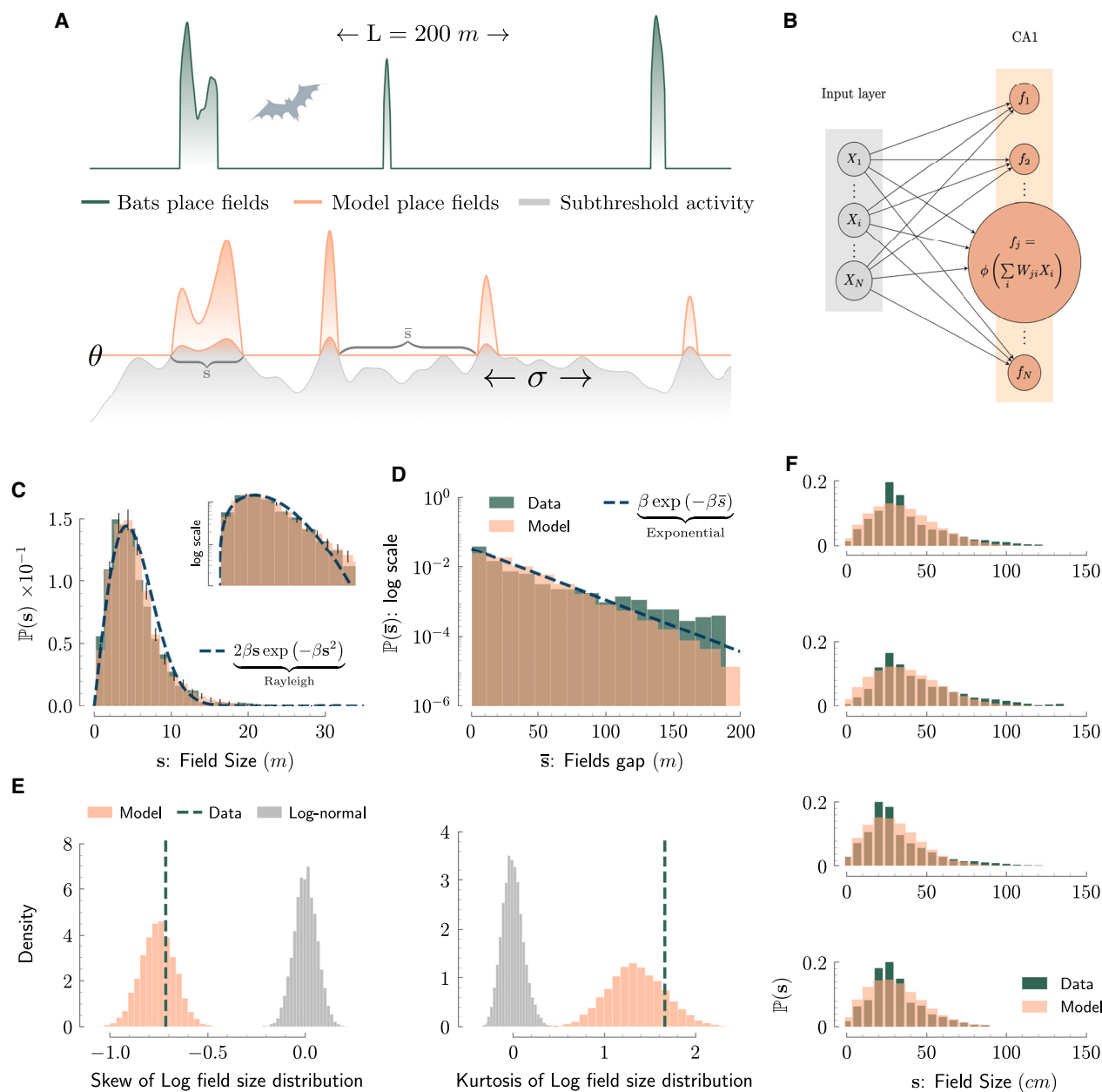


Figure 1. Field arrangements of 1D place fields are explained by the thresholded Gaussian process model

(A) Top: example of place fields measured in bats flying in a 1D 200-m-long tunnel.²¹ Bottom: firing rates in the model are generated by thresholding and rectifying a realization of a random Gaussian process (gray). In all figures, green represents experimental data, and orange represents the model.

(B) A simple neural architecture that gives rise to place fields that follow the thresholded Gaussian process model. Presynaptic inputs from many spatially selective neurons are summed with random synaptic weights to create the input to each CA1 place cell, followed by rectification.

(C) Field-size distribution of bats in the 1D tunnel compared against the field-size distribution predicted by the model. Dotted line: approximate analytical prediction of the field-size distribution, valid for high thresholds (Rayleigh distribution). Orange bars: precise predictions of the model, obtained from simulations (error bars: standard deviation across simulations, see STAR Methods). Here and in all subsequent bar plots, the overlap between green bars (experiment) and orange bars (model) is represented in brown. Inset: same data in a semi-logarithmic scale.

(D) Distribution of consecutive field gaps in the experiment compared against the distribution predicted by the model. Dashed line: analytical prediction.

(E) Skew (left) and kurtosis (right) of the log-field-size distribution, obtained from multiple simulations of the model and the best-fit heuristic log-normal distribution (gray). Each simulation is matched to the size of the experimental dataset. For the log-normal distribution, the empirical skew and kurtosis are expected to be close to zero. Dashed line: skew and kurtosis of the experimental distribution.

(F) Rayleigh distribution compared against place-field size distribution in four different mice, running in a 40-m-long virtual track.²⁴

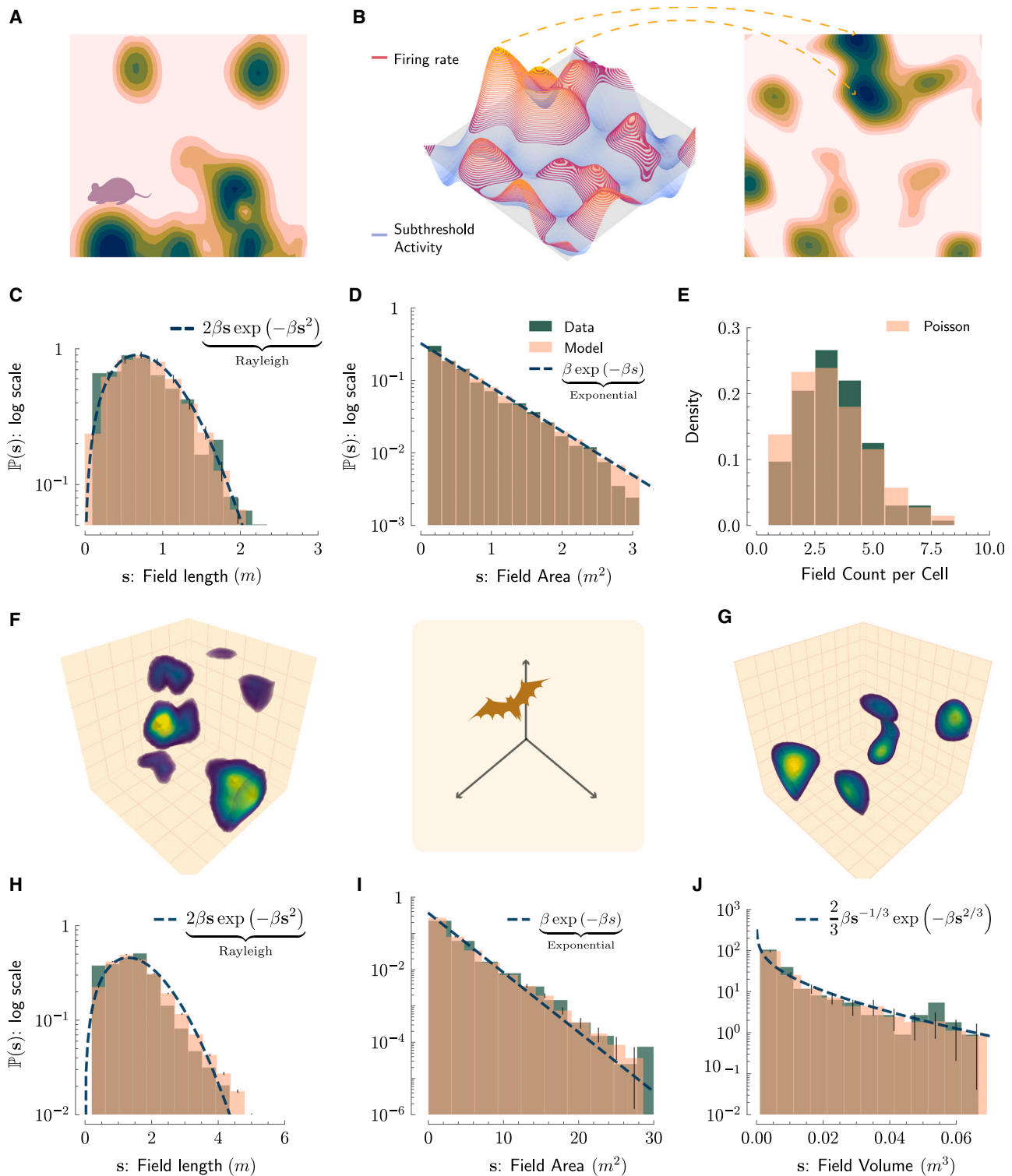


Figure 2. Field arrangements of place fields in 2D and 3D spaces are explained by the thresholded Gaussian process model

(A–E) Model fits to 2D data from rats.

(A) Sample spatial firing pattern of a CA1 cell recorded in a rat navigating in an 18.6 m² environment.²⁵

(B) Model: random Gaussian process in 2D (left) is thresholded to generate firing fields, yielding the cell's spatial firing pattern (right). For all depictions of distributions, the vertical scale is logarithmic and the color scheme for comparison between the model and the experiment is as in Figures 1C and 1D.

(C) Distribution of field lengths in 1D slices of 2D place fields (20 sample cells from Harland et al.²⁵).

(legend continued on next page)

two, and three dimensions. Building on our initial findings in 1D environments, we examined the extension of the model to 2D and 3D environments, where place fields are determined by excursion sets of a Gaussian process over the higher-dimensional space. As in the 1D case, the model generated multiple, heterogeneous fields, which were qualitatively similar to the ones observed experimentally (Figures 2A, 2B, 2F, and 2G).

Moreover, 1D slices through a multidimensional Gaussian process are themselves Gaussian processes. Hence, field sizes in 1D slices through multidimensional firing fields are expected to follow the same field-size distribution as in the 1D case. To test this prediction, we examined 1D slices through 2D place fields measured in rats foraging in a 18.6 m² arena (data from 20 cells recorded in Harland et al.²⁵) and found that they were well fitted by the Rayleigh distribution (Figure 2C). We further compared the model against a histogram of field areas from all cells (as reported in Harland et al.²⁵). The statistics precisely followed an exponential distribution as predicted by the model (Equation 1, $D = 2$; Figure 2D). Finally, field counts were well fitted by a Poisson distribution, as expected in the high-threshold regime in which field locations are independent (Figure 2E).

Next, we analyzed 3D place fields in bats navigating a large room of size 5.8 × 4.6 × 2.7 m³.²³ The model predicts that 2D slices through the 3D place fields should exhibit the same statistics as those of fields in 2D environments, since these slices are also realizations of a thresholded Gaussian process in two dimensions. The predictions for both 1D and 2D slices were verified by the data (Figures 2H and 2I), along with the distribution of the 3D field volumes (Figure 2J), which followed the prediction of Equation 1 with $D = 3$. In an alternative model, in which field volumes and peak firing rates were exactly matched to the empirical data, without the underlying Gaussian statistics, distributions of 1D and 2D slice sizes did not match the data (Figures S2A and S2B).

In summary, the statistics of field arrangements in 1D, 2D, and 3D were explained in both bats and rodents by the threshold crossings of Gaussian processes. In particular, the model explained the different field-size statistics that were observed across experiments in different dimensionalities (Equation 1; Figures 1C, 2D, and 2J). The generality of these results suggests that common mechanisms, shared across species and dimensionalities, underlie the structure of the hippocampal code for space.

Statistics of field shapes

The statistics of field sizes and gaps only reflect the statistics of field boundaries, which, in the model, correspond to threshold crossings. The model also makes quantitative predictions on the statistics of field shapes, defined as the firing rate's dependence on position within place fields (Figure 3A). These statistics

are determined by the properties of the Gaussian process in the threshold-crossing segments, yielding new theoretical predictions that can be tested in the experimental data. First, according to the model, the firing rate within a given place field may exhibit multiple local maxima, unlike in the classical picture of bell-shaped fields. The predicted distribution of the number of local maxima across fields was in close agreement with the empirical distribution of local maxima per field in bats flying in the 1D tunnel (Figure 3B). Second, the model predicts a positive correlation between the size of a field and its peak firing rate, with an approximate power law relation between these two quantities. The experimental data confirmed this prediction, in both bats flying in the 1D tunnel (Figure 3C) and bats flying in the 3D environment (Figure 3D). In both cases, the model not only predicted the qualitative relation between field size and peak firing rate but also accurately predicted the power law coefficient (Figures 3C and 3D). Third, in the model, the slope of the firing rate at the boundaries of 1D place fields follows a Rayleigh distribution.³⁵ This prediction was verified in the data (Figure 3E). Fourth, for bats flying in the 3D environment, the mean and Gaussian curvatures of iso-surfaces formed by the boundary of 3D place fields were distributed in agreement with the predictions of the model (Figures S2D and S2E). All the above predictions of the model were verified using the values of σ and θ that were previously inferred based only on field arrangements, independent of field shapes.

We next examined whether the model captures the topological properties of the place fields by using recent results on the Euler characteristic (EC) of excursion sets of Gaussian processes. The EC of the thresholded Gaussian process is a function of the number of connected components and holes of various dimensions (see STAR Methods) and has a universal dependence on the threshold, which is specific to Gaussian processes.³⁰ An analytical expression has been obtained for this quantity in terms of θ and σ . (This result has also been used to test for Gaussianity in the spatial structure of the cosmic background microwave radiation, as well as other natural processes.)^{36–39}

In the 1D case, the EC is simply equal to the number of connected components. The theory provides an exact expression for the EC curve (Equation 8), whose only dependence on the correlation function is through the correlation length σ . We tested this prediction for place fields by placing a varying threshold on the firing rates measured in bats flying in the 200-m-long tunnel, followed by evaluation of the EC of the rectified fields. The resulting EC curve (Figure 3F) was in excellent agreement with the analytical prediction using the previously obtained value of σ . While an increase in the threshold can be artificially applied to the measured rate maps, it is also possible to reduce the threshold by optogenetically depolarizing during behavior.

(D) Distribution of place-field areas from all cells in the experiment (experimental histogram adapted from Harland et al.²⁵).

(E) Distribution of field count per cell across the whole population of recorded cells (experimental histogram adapted from Harland et al.²⁵).

(F–J) Model fits to 3D data from bats.

(F) Sample 3D firing pattern recorded in a bat CA1 cell during flight in a 5.8 × 4.6 × 2.7 m³ room.²³

(G) Visualization of 3D receptive fields generated by the model.

(H) Distribution of field lengths in 1D slices of bat 3D place fields (analysis performed on all recorded cells in Yartsev and Ulanovsky²³).

(I) Distribution of field areas in 2D slices of the bat 3D place fields.

(J) Distribution of 3D field volumes in bats. As the dimension increases, the distribution becomes more concentrated around zero and develops a longer tail. All error bars represent standard deviation across simulations (see STAR Methods).

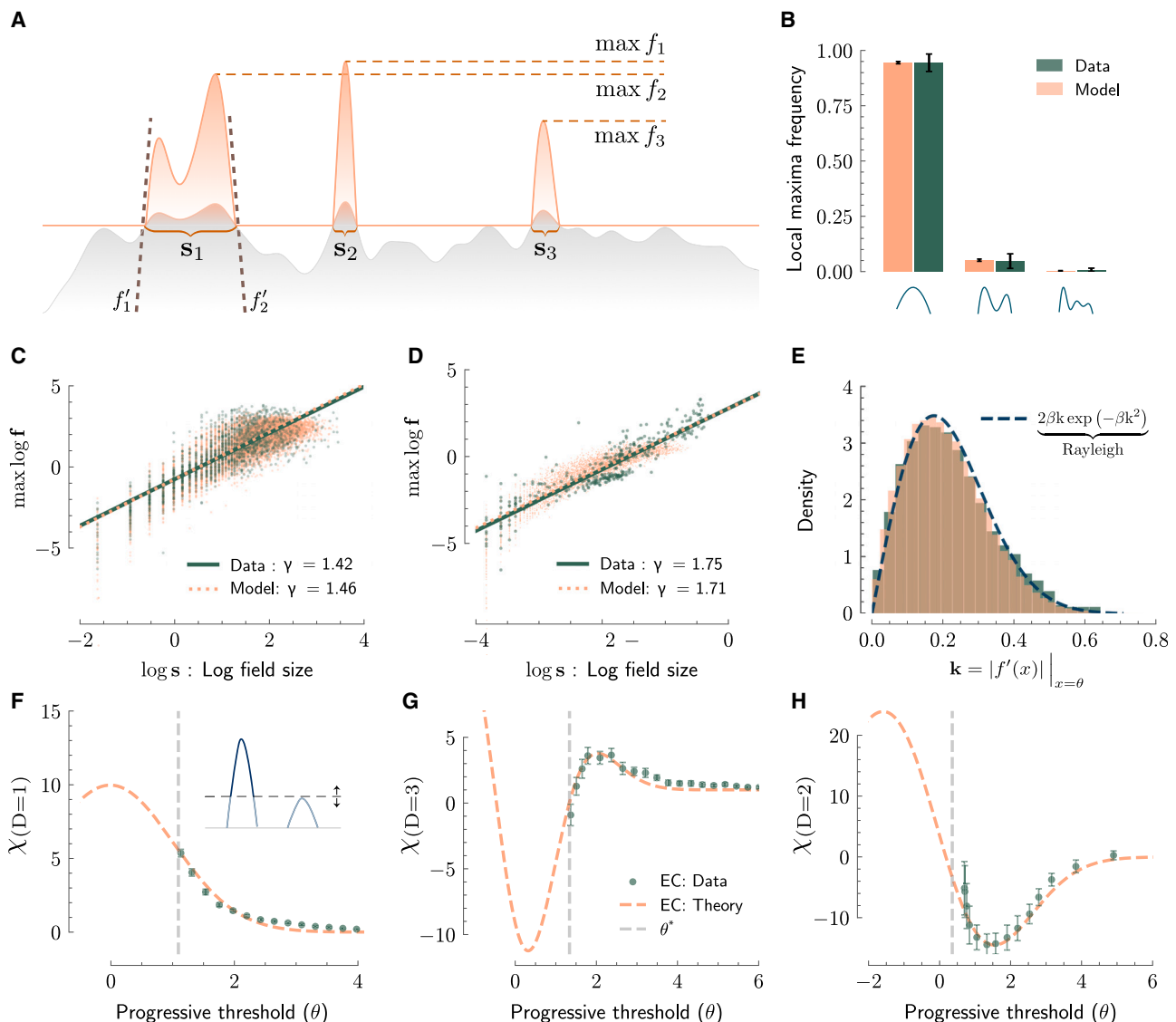


Figure 3. Field-shape-related statistics of place fields are predicted by the thresholded Gaussian process model across dimensions and species

(A) Visualization of various field-shape-related measurement in the case of 1D fields.

(B) Distribution of the number of peaks in the place fields of bats in the 200-m-long tunnel²¹ compared with the model. Error bars: standard deviation across simulations (model) and standard deviation across a range of smoothing parameters (data, see STAR Methods).

(C) Joint distribution of log receptive field size (in m) and log maximal firing rate (in Hz) obtained from the model (orange) and from experimentally measured bat place fields in the 200-m-long tunnel (green). Lines: power law relationship between the two quantities in the model and experiment. The coefficient γ is the slope of the best linear fit to the distribution and represents the power law coefficient. The γ intercept of the fit is matched between data and model to infer the linear gain factor.

(D) Same as in (A), for place fields recorded in bats flying in the $5.8 \times 4.6 \times 2.7 \text{ m}^3$ room.²³ Field size is the cubic root of the place-field volume (in m).

(E) Absolute value of field slopes at their boundaries follow a Rayleigh distribution.

(F) EC (χ) as a function of extra thresholding of the place fields of bats in the 200-m-long tunnel (dashed line: theoretical prediction, dots: data). Inset illustrates how adding an extra threshold (horizontal dashed line vs. solid line) can affect the number of fields.

(G) Similar plot as (D) for bats in 3D environment.²³

(H) Same as (F) and (G) in the 18.6 m^2 2D environment (using 20 sample cells from rats²⁵).

Error bars in (F) and (G) represent standard deviation across neurons.

Recent results from such experiments⁴⁰ are in qualitative agreement with the threshold-crossing model since the stimulation led to the emergence of unmasked place fields.

In dimensions higher than one, the EC depends not only on the number of connected components but also on the number of holes of various dimensions within the fields. This results in a

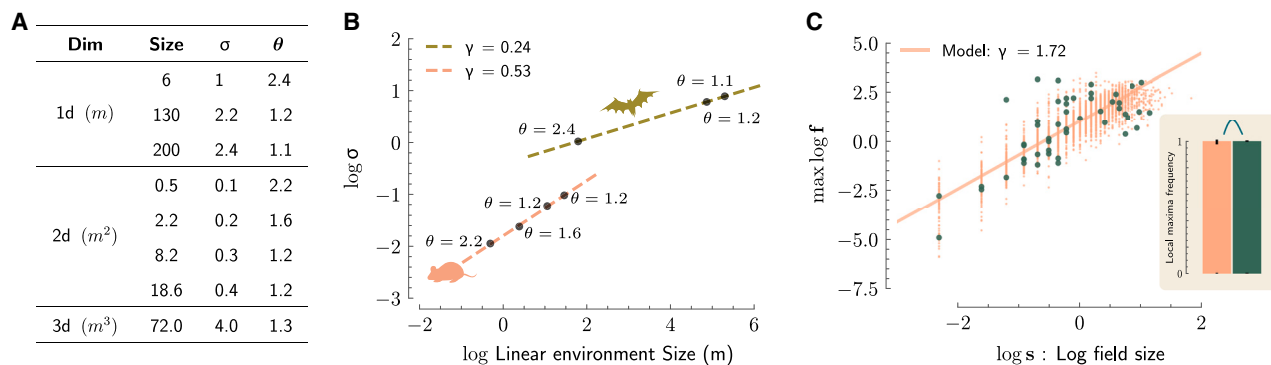


Figure 4. Parameter trends across experiments explain differences in heterogeneity

(A) Summary of parameters for the various experiments in spaces of varying dimension: 1D, bats flying in a tunnel; 2D, rats navigating a large open arena; and 3D, bats flying in a room.

(B) Linear scaling of $\log \sigma$ with the logarithm of the linear environment size in 1D (bats) and 2D (rats).

(C) Joint distribution of log receptive field size, s (in m) and log maximal firing rate, f (in Hz), plotted as in Figure 3C, but in the 6-m-long tunnel. In comparison with the 200-m-long tunnel (Figure 3C), the exponent characterizing the relationship between field size and maximal firing rate is closer to 2, indicating that field shapes in the 6-m-long tunnel are more stereotyped than in the 200-m-long tunnel, as expected due to the higher threshold. Inset: fields with multiple peaks are practically absent, as predicted (compare with Figure 3B).

In (B) and (C), γ is the slope of the best linear fit to the data in logarithmic scales.

non-monotonic dependence of the EC curve on the threshold due to the emergence and disappearance of holes. Here, too, the theory provides an exact analytical expression (Equation 10) for the expected EC curve. We tested this prediction on all measured firing fields in bats flying in 3D.²³ The empirical EC curve was non-monotonic as predicted and precisely followed the analytical expression (Figure 3G) using the previously obtained value of σ . This agreement is specific to the Gaussian threshold-crossing model and was not reproduced by an alternative model with matching joint statistics of field volumes and peak rates (Figure S2C). We also tested the EC curve prediction in 2D on 20 cells from rats recorded in the 2D megaspace²⁵ and obtained excellent agreement with the analytical prediction (Equation 9; Figure 3H).

Parameter variation across experiments

Our analyses of data from 1D and 2D environments of varying sizes allow us to identify qualitative trends across datasets, as a function of the environment size (Figures 4A and 4B). We observe that the correlation length, σ , increases sublinearly with the size of the environment, and the normalized threshold, θ , decreases with the size of the environment (Figure 4B). Together, these trends induce an increase in the average number of fields with the size of the environment, consistent with the observation of single firing fields per cell in small environments and a sublinear increase in their sizes (Figure S4). It is also interesting to consider the number of fields divided by the environment size, to which we refer as the “average propensity.”²⁴ The sublinear increase of σ and the decrease of θ have opposing influences on this quantity. Consequently, the average propensity varies only weakly with environment size (Figure S3).

While the two parameters jointly affect the field counts and the field sizes, the normalized threshold also influences the heterogeneity in field shapes (Figure 5A). With increasing θ , the fraction of multi-peaked fields decreases and the exponent of the power

law relationship between the maximum firing rate and the field size approaches 2, as expected for parabolic fields³⁰ (Figures 5B and 5C). Thus, the model predicts a greater stereotypy of field shapes with an increasing normalized threshold. As a result, and following the trend seen in the data (Figures 4A and 4B), field shapes are predicted to be more stereotyped in smaller environments than in larger ones (Figures 5A–5C). We tested this prediction on place fields measured in bats flying in a short, 6-m-long tunnel.²¹ As predicted, field shapes were more stereotyped than in the 200-m-long tunnel, with statistics that quantitatively agreed with the model (Figures 4C and 5D). In summary, the model accounts for the different characteristics of CA1 spatial selectivity across environmental scales, ranging from early experiments in small environments, where place cells typically had a single, bell-shaped firing field,^{8,22} to more recent experiments in large environments, where place cells had multiple firing fields with heterogeneous shapes.^{10,21,24,25}

Each one of the experimental datasets is well explained by a single choice of the values of σ and θ , yet some variation in these parameters across cells and anatomical regions is to be expected. For the length scale σ , our choice of a single value is motivated by the fact that recordings were tightly localized anatomically in dorsal CA1.^{21,23,25,41} In the data from bats flying in the 1D tunnel, the number of place fields expressed by individual neurons, as well as their sizes, were observed to be correlated in the two flying directions, despite remapping of the fields.²¹ We hypothesized that this correlation arises from variability in the threshold, θ . In agreement with this hypothesis, an extended model with a distribution of thresholds accounted for the empirical correlations, while remaining compatible with the distributions of field sizes and gaps obtained from the basic model (Figure S4). A distribution of thresholds may arise from a variation in intrinsic cell properties or from variation in the inhibitory inputs to place cells and is consistent with the recent observation that propensity varies across cells also in rodents.²⁴

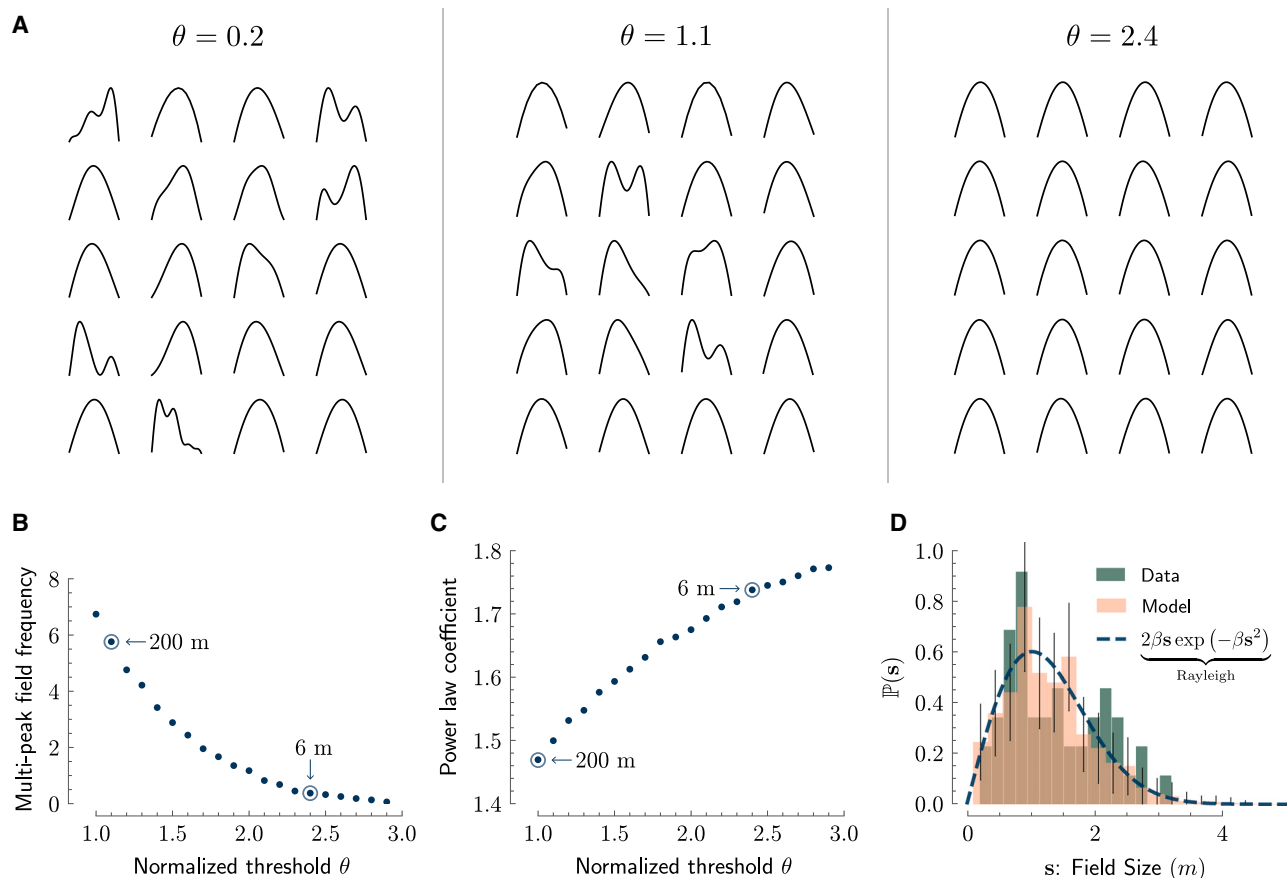


Figure 5. Field shape as a function of threshold

(A) Visualization of randomly selected sample field shapes generated by the model ranging from low- to high-threshold regime (left to right, the intermediate value of $\theta = 1.1$ is the normalized threshold inferred from recordings in the 200-m-long tunnel and the largest value $\theta = 2.4$ is the normalized threshold inferred from recordings in the 6-m-long tunnel).

(B) Predicted percentage of cells with more than one local peak, as a function of the normalized threshold.

(C) Predicted logarithmic slope of field width vs. max. firing rate, as a function of the normalized threshold. It can be shown that in the limit of very high threshold, fields are parabolic and the logarithmic slope approaches 2.³⁰

(D) Field widths in bats flying in a 6-m-long tunnel follow a Rayleigh distribution despite more stereotypy in field shapes as predicted (40 recorded cells; peak count in the figure is 8; the distributions have no significant difference; Kolmogorov-Smirnov [K-S] test: p value = 0.6, K-S stat = 0.1). Error bar represents standard deviation across simulations (see STAR Methods).

DISCUSSION

The classical view of spatial coding in the hippocampus has been challenged by the recent discoveries of heterogeneous and distributed CA1 response patterns in large environments. It was unknown whether a unified mathematical framework can encompass the diverse statistics of these irregular responses. Here, we identified such a framework, in which place fields are obtained from the threshold crossings of a spatially fluctuating random process. The underlying randomness of the Gaussian process enabled the model to explain in quantitative detail the variability in the size of place fields and their spatial arrangement, as well as a range of geometric and topological properties associated with the heterogeneous field shapes. Indeed, when constrained only on the mean field size and count, the model precisely captured the statistics of shape variability, which, in principle, could have had an independent origin. This surprising result sug-

gests that the different degrees of heterogeneity, observed across experiments, arise as a byproduct of mechanisms that regulate the field frequency and size. With only two parameters, the model accounts for the seemingly diverse statistics observed in rodents and bats and in environments of varying sizes and dimensionalities. Therefore, the model allows for direct comparison between the statistics observed in the different experiments and suggests that they arise from common underlying principles. Specifically, our analysis of the experimental data indicates that there are systematic differences in the statistics of place fields in environments of different size: field sizes increase with environment size, propensity decreases, and field shapes become more heterogeneous. Consequently, place fields in small environments differ from those observed in a similarly sized portion of a larger environment.

A Gaussian process is the random process that maximizes entropy under constraints on the mean and the spatial covariance function. Hence, the model invokes minimal assumptions other

than a specification of a place cell's underlying input correlation structure.^{31,42} At the same time, random Gaussian processes can generate a highly efficient coding scheme, with capacity that increases exponentially with the number of neurons,²⁹ and, therefore, the structure of tuning curves in CA1 might be shaped to optimize coding efficiency. Another possibility is that the structure of neural responses in CA1 is beneficial for the maintenance and retrieval of associative memory,²⁷ since CA1 is part of a larger system involved in this function.⁴³ It will be interesting to examine theoretically whether it is possible to obtain a coherent understanding of the trends seen in the parameters across the different geometries (Figure 4B) in light of a theory of efficient coding²⁹ or of optimal storage capacity.

The success of the parsimonious model introduced here invites an exploration into its possible mechanistic origin. The apparent stereotypy of place fields in small environments has motivated a view of hippocampal spatial selectivity as arising from highly organized synaptic connectivity.^{44–46} However, the irregular CA1 firing patterns in large environments are inconsistent with this view. Our results indicate that CA1 place fields, in small and large environments alike, are compatible with predominantly random synaptic connections into CA1. This is because random projections from a sufficiently large number of spatially selective cells necessarily produce an input to each CA1 cell that varies spatially as a realization of a random Gaussian process, due to the multivariate central limit theorem (see [Methods S2](#)). Gaussian statistics arise robustly from random synaptic projections without the necessity to invoke highly specific assumptions on the structure of spatial responses in the input layer. These can range from regular tuning curves that tile the environment to heterogeneous and irregular tuning curves with a distance-dependent covariance function (see [Methods S2](#)). Hence, the broad success of the model across datasets and statistics suggests that randomness, rather than specific design, governs the synaptic organization of inputs to CA1.

Randomness in the connectivity might be associated with a fairly fixed anatomical structure, potentially influenced by the developmental trajectory of neurons and their embedding in the network.⁴⁷ Yet, the strength of synaptic projections between CA3 and CA1 is known to be plastic. Specifically, recent experiments indicate that a dominant mode of plasticity in CA1 involves randomly occurring modifications of synaptic projections from CA3, triggered by inputs from the entorhinal cortex.^{48–50} The randomness of the synaptic weights may be related to the cumulative effect of many such non-Hebbian plasticity events. Since all the measurements analyzed in this work were collected in highly familiar and uniform environments, it is possible that the stationarity of the underlying Gaussian processes is specific to such conditions. We speculate that more intricate features of the place-cell code arise under richer behavioral conditions via activity-dependent plasticity mechanisms acting on top of a scaffold of random synaptic weights.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Yoram Burak (yoram.burak@elsc.huji.ac.il).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- This paper analyzes existing, publicly available data, accessible at <https://doi.org/10.5281/zenodo.46467270> and <http://amygdala.psychdept.arizona.edu/lab.html>.
- All original code is publicly available at <https://doi.org/10.5281/zenodo.14650880>.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

N.M. and Y.B. conceptualized the research and developed the methodologies, with major inputs from R.A.d.S. All authors interpreted the results. N.M. developed the code for simulation and data analysis, performed simulations, and analyzed the data. N.M. and Y.B. wrote the first draft of the manuscript with major inputs from R.A.d.S. All authors participated in writing and editing of the manuscript. Y.B. supervised the project and acquired funding.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Datasets	Eliav et al. ²¹	https://doi.org/10.5281/zenodo.4646727
Datasets	Harland et al. ²⁵	http://amygdala.psychdept.arizona.edu/lab.html
Software and algorithms		
Python 3.11	Python Software Foundation	https://www.python.org/
Code for this paper	https://doi.org/10.5281/zenodo.14650880	https://github.com/5183nischal/gp-model

METHOD DETAILS

Mathematical notation

Throughout the main text, STAR Methods section, and supplemental information, we use the mathematical notation defined below:

Symbol	Description
L	Length of one-dimensional (1D) environment
D	Dimension of space
L^D	Volume of environment for $D > 1$
x	Spatial coordinate
$\mathcal{GP}$	Gaussian process
$h(x)$	Gaussian process input to CA1 cell
$r(x - x')$	Covariance function of $h(x)$
$r(0)$	Variance of $h(x)$
$-r''(0)$	Variance of $h'(x)$, the derivative process
θ	Normalized threshold
σ	Correlation length
$f(x)$	Firing rate of CA1 cells in the model
$\phi(\cdot)$	Gaussian cumulative density function
N	Field count per cell
s	Field size
$\bar{s}$	Field gap
χ	Euler characteristic

Model details

Place fields are modeled as rectified realizations of a stationary Gaussian process over D dimensional space ($x \in \mathbb{R}^D$) with 0 mean and stationary covariance function $r(x - x')$,

$$h(x) \sim \mathcal{GP}[0, r(x - x')]. \quad (\text{Equation 2})$$

A sample $h(x)$ from the Gaussian process can be thought of as input to a CA1 place cell. The input is subsequently thresholded at a value $\theta\sqrt{r(0)}$ to generate the cell's firing rate, which is proportional to $f(x)$, defined as

$$f(x) = \max(h(x) - \theta\sqrt{r(0)}, 0), \quad (\text{Equation 3})$$

where θ is the normalized threshold, equal to the ratio between the threshold and the standard deviation of the process. The correlation length of the Gaussian process is defined as³⁰:

$$\sigma = \sqrt{\frac{r(0)}{r''(0)}}. \quad (\text{Equation 4})$$

Thus, σ is equal to the ratio between the standard deviation of the process $\sqrt{r(0)}$ and the standard deviation of its derivative $\sqrt{-r''(0)}$.³⁰ For simulations of the Gaussian process, we used a transitionally invariant Gaussian correlation function.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data processing

Place field firing rate maps (post place field identification) in bats in 1D²¹ and 3D²³ environments were obtained and rectified at ≈ 0.5 Hz for denoising. In the 3D case, firing rates also went through a median filtering step using *scipy.ndimage* Python library with minimal (nearest neighbor) kernel of size 2 (place fields resolution = 10 cm on each axis) to eliminate salt-and-pepper noise while preserving field boundaries.

Parameter inference in 1D environments

The two parameters σ and θ were inferred by matching the means of the field size and the field gap to those in the model via simulation in the dataset from bats in the 200 m²¹ (Figures 1C, 1D, 4A, and 4B). In the dataset from bats in the 6-m-long tunnel (Figure 5D), most cells had a single field and parameters were inferred by matching the mean field size and the mean fraction of the environment in which place cells were active. For the dataset from bats in the 130-m-long tunnel and dataset from rats in the 48-m-long maze, mean field size and mean field count were matched to infer the parameters^{10,21} (Figures S1, 4A, and 4B). For rats in a 40-m-long virtual track²⁴ (Figure 1F), where only a histogram of field sizes was available, we directly fitted the parameter β of the approximate Rayleigh distribution (Equation 1). Parameters matched through simulation yielded excellent agreement with the analytical formulas in the 1D case (see Methods S3, Equations S21 and S22).

Parameter inference in 2D environments

For rats in the 2D environments²⁵ (Figures 1C–1E, 4A, and 4B), parameters were inferred by fitting the model to the mean field count and mean field size reported in Harland et al.²⁵ When matching the mean count, we took into account the systematic exclusion of cells with zero fields from experimental observations by fitting the model to the distribution of field counts, conditioned on the existence of at least one field. For the Poisson distribution (valid in the high threshold regime), the mean $\langle N \rangle_{\text{experiment}}$ of the conditioned distribution, which is matched to the experimental data, is then

$$\langle N \rangle_{\text{experiment}} = \frac{\langle N \rangle}{1 - \exp(-\langle N \rangle)} \quad (\text{Equation 5})$$

where $\langle N \rangle$ is the unconditioned mean. Analysis of the Euler Characteristic in the large 2D environment²⁵ (Figure 2H) was performed on a sample of firing rate maps from 20 cells shared with us by the authors of the manuscript. All other 2D results are based on statistics from the full dataset, as reported in Harland et al.²⁵

Parameter inference in 3D environment

For bats in a 3D environment²³ (Figures 2H–2J and 4A) the parameters σ and θ were inferred by matching the mean field volume and the mean fraction of the volume in which place cells were active, based on the full rate maps.

Fields in 1D

Fields were defined as connected components of non-zero activity, and gaps as connected components of zero activity (Figures 1C, 1D, 2B, and 2E). When generating histograms based on the model (Figures 1C, 2C, 2H–2J, and 4C), errors bars were obtained by performing 10^4 simulations of random fields using the model, each matching the number of cells to the data. Field size distributions from model simulations in 1D were compared against the best fit log-normal distribution by first generating a kernel density estimate of the field size distribution from the simulations (*scipy.stats.gaussian_kde*). The model has one less degree of freedom than the two-parameter log-normal distribution, since constraining the model parameters to also fit the gap size distribution reduces the parameter space to one dimension. Nevertheless, the log likelihood test indicated that the model fits the data better than the log-normal distribution ($\Delta \text{LLR} = -50.3 < 0$). Differences in model complexity aware log likelihoods were measured using the Bayesian Information (BIC) and Akaike Information (AIC) criteria, yielding $\Delta \text{BIC} = -109.1 < 0$ and $\Delta \text{AIC} = -104.5 < 0$, which implies that the data fit the model better than log-normal.

Kurtosis and skew

Kurtosis, quantifying the heaviness of distribution tails, was defined as

$$\text{Kurtosis} = \frac{m_4}{m_2^2} - 3, \quad (\text{Equation 6})$$

where m_4 and m_2 are the fourth and second moments of the distribution of log-field sizes, respectively. Skew, quantifying the asymmetry of the distribution was defined as

$$\text{Skew} = \frac{m_3}{m_2^{3/2}} \quad (\text{Equation 7})$$

where m_3 and m_2 are the third and second moments of the log-field-size distribution, respectively. Both measures vanish for a log-normal distribution, since in that case the logarithm of field sizes is normally distributed. The effect of rectification as part of data processing is explored in Figure S5A.

Fields in 3D

Field statistics in 3D were measured from the firing rate data from Yartsev and Ulanovsky.²³ To measure field volumes, connected components in the 3D volumetric fields were identified using the Python function `scipy.ndimage.label`, which identifies and labels connected regions in N-dimensional arrays. Fields with volume larger than 3 standard deviations from the mean were removed as outliers, as they are likely to result from merging of two or more fields. These large outliers were also removed when analysing the slices of the 3D fields (Figures 2D–2F).

Number of local peaks within a field

Maxima in each field from 1D data were identified using the Python function `scipy.signal.find_peaks`, which finds local peaks in a 1D array. Incomplete fields at boundaries were removed. To filter out spurious maximas in the data arising from noisy assessment of the firing rate, we first applied a Gaussian smoothing with a standard deviation ranging from 0.5 to 1 m in the case of bats flying in a 200-m-long tunnel, and Gaussian smoothing with a standard deviation ranging from 0.3 to 0.6 m in the case of bats flying in a 6-m-long tunnel. Local peak count distribution was calculated for the smoothed fields in each case and mean and standard deviation are reported. The error bars in the model represent the standard deviation across independent simulations (Figures 3B and 4C).

Maximum firing rate vs. field width

Maximum firing rates in Figures 3C and 3D were defined as the maximum rate observed in each field within its connected region. In the 1D data (Figure 3C), fields were rectified by a small threshold of 1 Hz to eliminate spurious measurements arising from regions in which the firing rate measurement is unreliable (note that the Gaussian threshold crossing model is applicable for the rectified data). The power law coefficient extracted from the data was insensitive to the choice of the threshold above this value (Figure S5B). The y intercept of the linear fit on the log-log plot was matched to eliminate the effect of the linear gain factor.

Boundary slopes

Field boundary slopes were extracted by fitting a line between the two consecutive samples of the firing rate (separated by 0.1 m) at the boundary of each rectified field, and measuring its slope (Figure 3E).

Euler characteristics

The Euler characteristic (EC) is a topological invariant that can be expressed as the alternating sum of Betti numbers. The 0th order Betti number is equal to the number of connected components in a topological space, and the n th order Betti number is equal to the number of n -dimensional holes. The EC χ of a Gaussian process in D dimensions has a universal form, insensitive to the structure of the underlying covariance function (under mild regularity requirements³⁰):

$$D = 1 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)}{2\pi} \right] e^{-\frac{\rho^2}{2}} + (1 - \phi(\theta)), \quad (\text{Equation 8})$$

$$D = 2 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)^2}{(2\pi)^{3/2}} \theta + \frac{2(L/\sigma)}{2\pi} \right] e^{-\frac{\rho^2}{2}} + (1 - \phi(\theta)), \quad (\text{Equation 9})$$

$$D = 3 \Rightarrow \langle \chi \rangle = \left[\frac{(L/\sigma)^3}{(2\pi)^2} (\theta^2 - 1) + \frac{3(L/\sigma)^2}{(2\pi)^{3/2}} \theta + \frac{3(L/\sigma)}{2\pi} \right] e^{-\frac{\rho^2}{2}} + (1 - \phi(\theta)), \quad (\text{Equation 10})$$

where ϕ is the Gaussian CDF function. Empirical Euler characteristics (EC) were measured using the *Gudhi* python library by taking the alternating sum of Betti numbers. The mean maximum firing rate was normalized in the data to match that of the model to determine the linear gain value before calculating the EC.

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Supplemental information

**Universal statistics of hippocampal place fields
across species and dimensionalities**

Nischal Mainali, Rava Azeredo da Silveira, and Yoram Burak

Supplementary materials

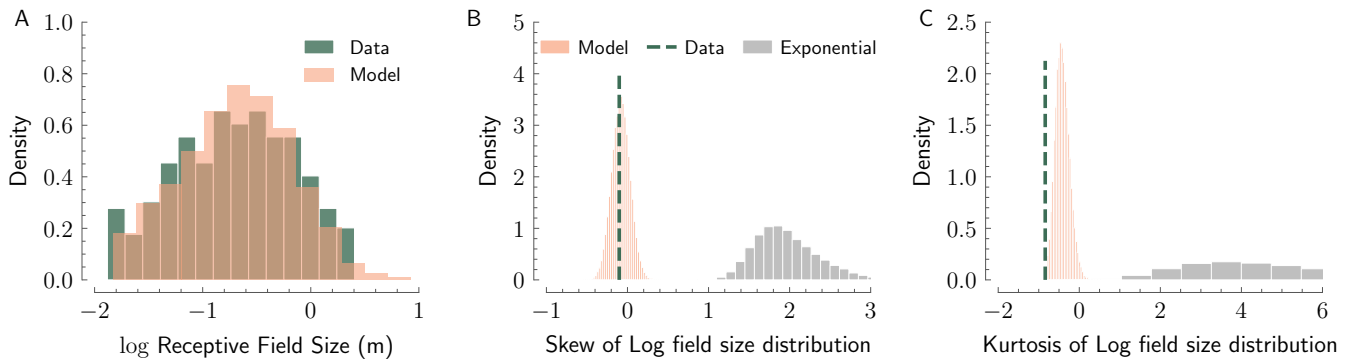


Figure S1. Field arrangements from rats in 1D environments are explained by the model. Related to Figure 1.

(A) Distribution of (log) field sizes measured in rats navigating a 48 m-long 1d maze ^{S1} along with distribution arising from the model ($\sigma = 0.34$, $\theta = 1.8$), fitted to match mean field size and count observed in the data. Fields in data larger than three times the standard deviation were removed as outliers. The empirical distribution matches the model better than the heuristic exponential distribution proposed in ^{S2} (likelihood test: $\Delta \log \text{Likelihood} = -83.1 < 0$).

(B-C) Skew/Kurtosis distribution of log field size obtained from multiple simulations of the model (orange) and the heuristic exponential distribution (gray) compared against that extracted from the experimental data (vertical green dashed line). The empirical fields were required to be larger than 15 cm in ^{S1}. We applied the same cutoff on the fields from the model to match the distribution obtained from the data. Note that this affects the distribution of skew and kurtosis.

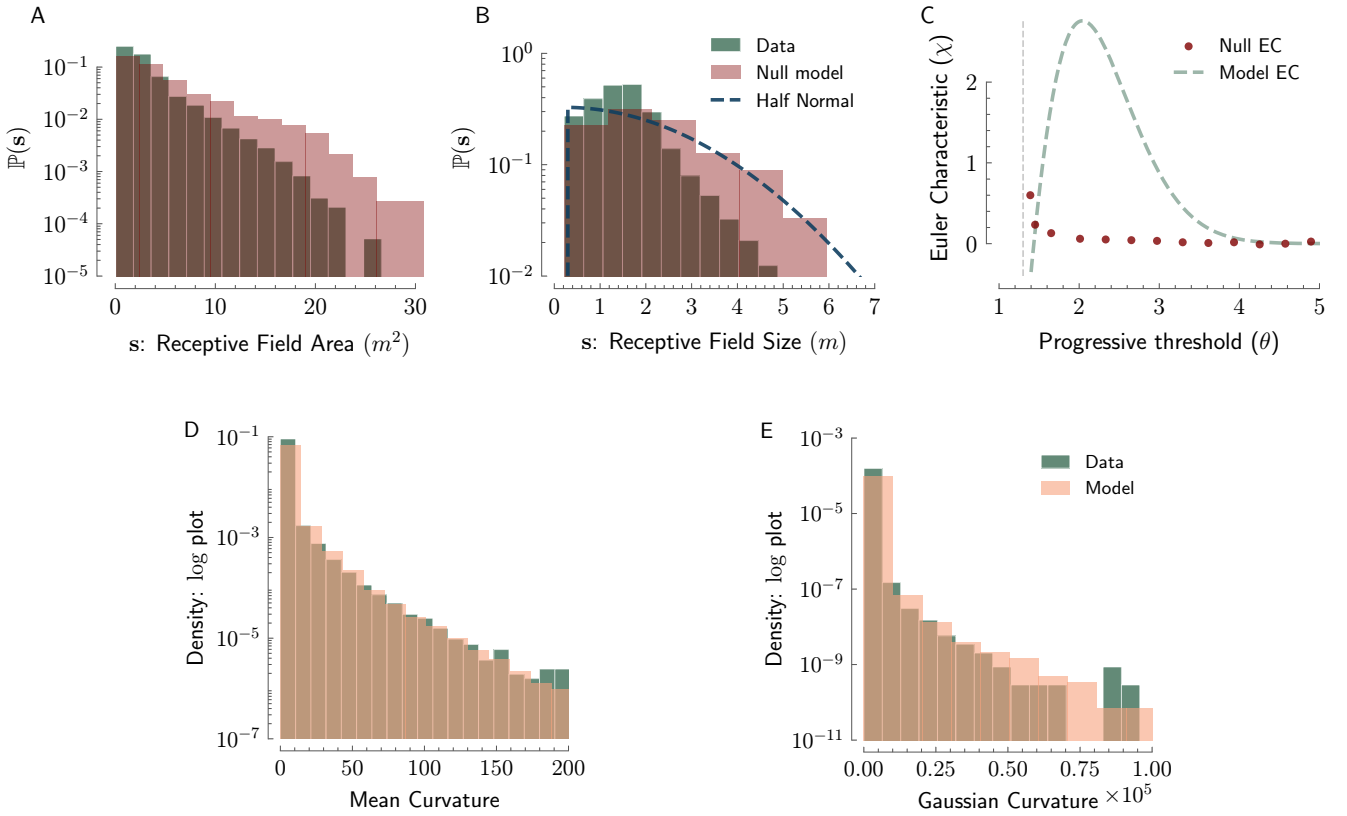


Figure S2. Further analysis of 3d place fields: comparison to alternative model with spherical fields and curvature of field iso-surfaces. Related to Figure 2 and Figure 3.

(A/B/C) Alternative model for comparison against the thresholded Gaussian process model in 3d. Firing fields in the alternative model are spherical, while the joint distribution of field volumes and maximum-firing rates exactly matches the empirical distribution. More precisely, for each field in the data, we created a corresponding field in the model with spherical support and matching volume. These fields are not allowed to overlap but are otherwise randomly placed since the measurement of field sizes in the slices and the expected Euler characteristics are independent of the field placements. The firing rate profile for each of these fields was modeled as a quadratic function with amplitude exactly matching that of the field in data.

(A) Discrepancy between areas of 2d slices through 3d fields in data and in the alternative model (logarithmic vertical scale).

(B) Discrepancy between length of 1d slices through 3d fields in data and in the alternative model (logarithmic vertical scale). In the null model, the radius of each spherical field is distributed as Rayleigh distribution, which implies that the length of active portions of 1d slices follows a half-normal distribution.

(C) Comparison between Euler characteristics of the thresholded Gaussian process model (matching the empirical data, Fig 3G), and Euler characteristics arising from the alternative model.

(D) Distribution of mean curvature and (E) Gaussian curvature of 3d place field boundaries in data and model (logarithmic vertical scale). The boundary surface for each field was first identified and represented as a two-dimensional array Z , where each point (x, y, z) of the surface was mapped to position (x, y) in the array containing the z -value of the surface points. The mean and Gaussian curvatures of the isosurface representing the boundaries of the 3D fields were then

calculated^{S3} using the following operators. Gaussian curvature (K) is defined as:

$$K = \frac{\partial_{xx}\partial_{yy} - \partial_{xy}^2}{(1 + \partial_x^2 + \partial_y^2)^2}$$

and mean curvature (H) is defined as:

$$H = \frac{(1 + \partial_x^2)\partial_{yy} + (1 + \partial_y^2)\partial_{xx} - 2\partial_x\partial_y\partial_{xy}}{2(1 + \partial_x^2 + \partial_y^2)^{3/2}}$$

where ∂_x , ∂_y , ∂_{xx} , ∂_{yy} , and ∂_{xy} are the first and second-order partial derivatives of the surface represented by array Z with respect to x and y . The partial derivatives were computed using the `numpy.gradient` function, which calculates the numerical gradients of the input array.

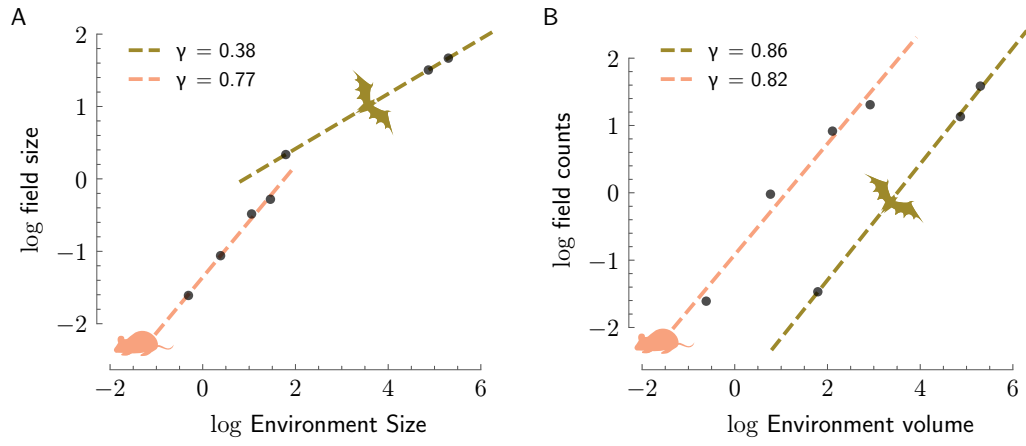


Figure S3. Trends in field statistics as a function of environment size. Related to Figure 4.

(A) Linear scaling of logarithm of linear field sizes (in m) with the logarithm of the linear environment size (in m) in 1d (bats) ^{S4} and 2d (rats) ^{S5}. In both panels, γ is the slope of the best linear fit to the data, in logarithmic scales.

(B) Linear scaling of logarithm of field counts with the logarithm of the linear environment volume (length in 1d and area in 2d) in 1d (bats) ^{S4} and 2d (rats) ^{S5}. The average propensity, defined as the average field count divided by the size of the environment (length in 1d and area in 2d) decreases weakly with environment size, exhibiting an approximate power-law dependence on the size of the environment, with an exponent $\gamma - 1$, equal approximately to -0.14 in 1d and to -0.18 in 2d.

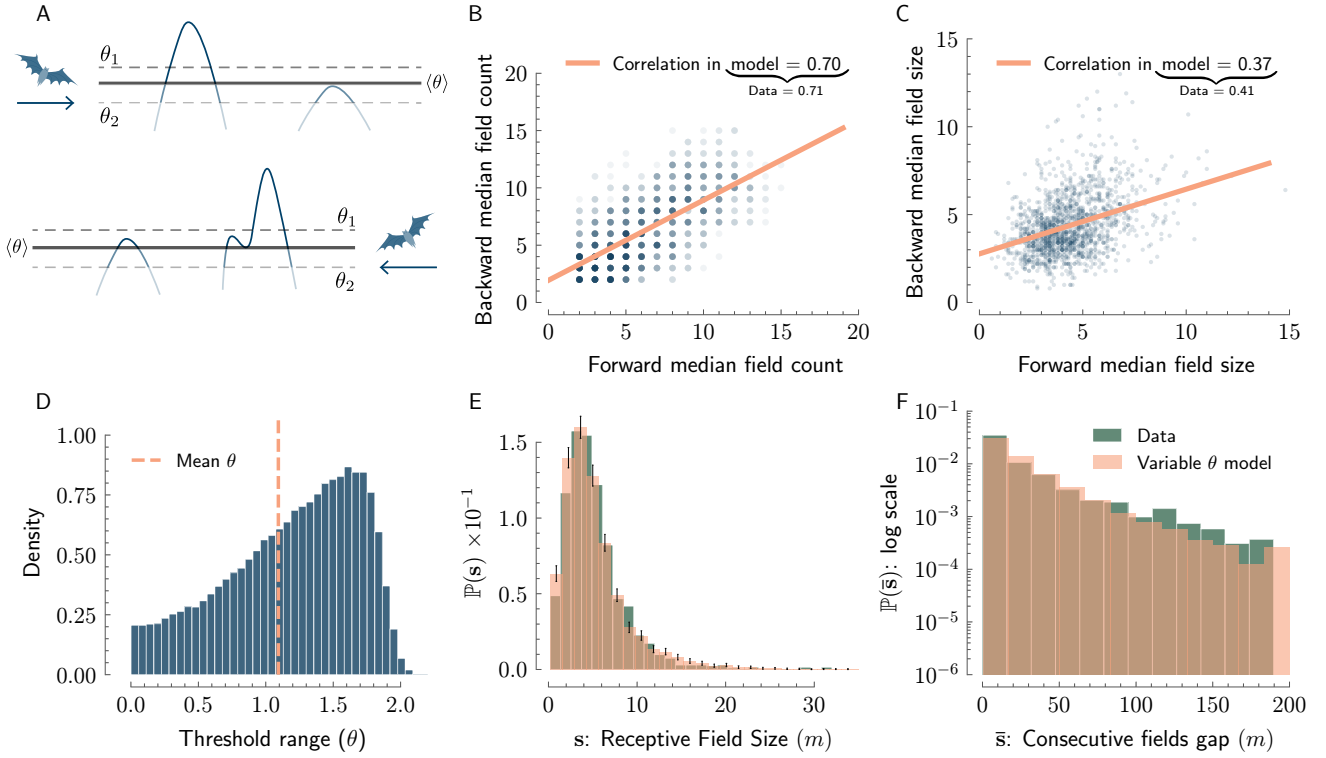


Figure S4. Threshold noise explains correlation in field statistics between flight directions of bats in a 1d environment. Related to Figure 1 and Figure 5.

(A) Illustration of how variability in the normalized threshold across cells leads to correlated field counts and field sizes in the two flight directions along the 1d tunnel. Neurons with lower threshold will have more place fields on average, when the bat flies along both directions of the tunnel. Likewise, neurons with lower threshold will have larger field size on average, when the bat flies along both directions of the tunnel.

(B) Scatter plot (blue) and correlation of field counts obtained from the model in two different samples (representing forward and backwards directions) of fields with fixed noisy threshold. With a distribution of thresholds, the model matches the empirical correlation between field counts in bats flying in the forward and backward directions (parameters chosen as explained in D).

(C) Scatter plot (blue) and correlation of median field size per cell in two different samples of fields with fixed noisy threshold. With a distribution of thresholds, the model matches the empirical correlation between field sizes in the two flying directions, while simultaneously matching the correlation in field counts (parameters chosen as explained in D).

(D) The distribution of θ in the extended model with variability in the threshold across cells. θ is sampled from a skew-normal distribution conditioned to be positive, with mean matching the single inferred θ in case of bats in 1d tunnel. The skew normal distribution has two remaining parameters after fixing the mean, which were set to match the correlation between median field count and median field size as reported in ^{S4}.

(E) Field size distribution under the variable threshold model matches the distribution of field sizes in the data.

(F) Field gap distribution in the model matches that of the data and further explains (without additional fitting) the slightly heavier tail in the empirical distribution in comparison with the fit to an exponential distribution (compare with Fig. 1D).

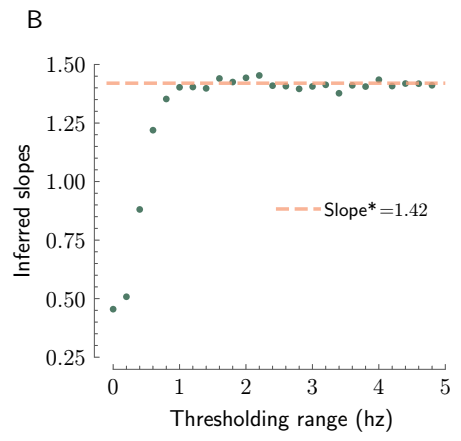
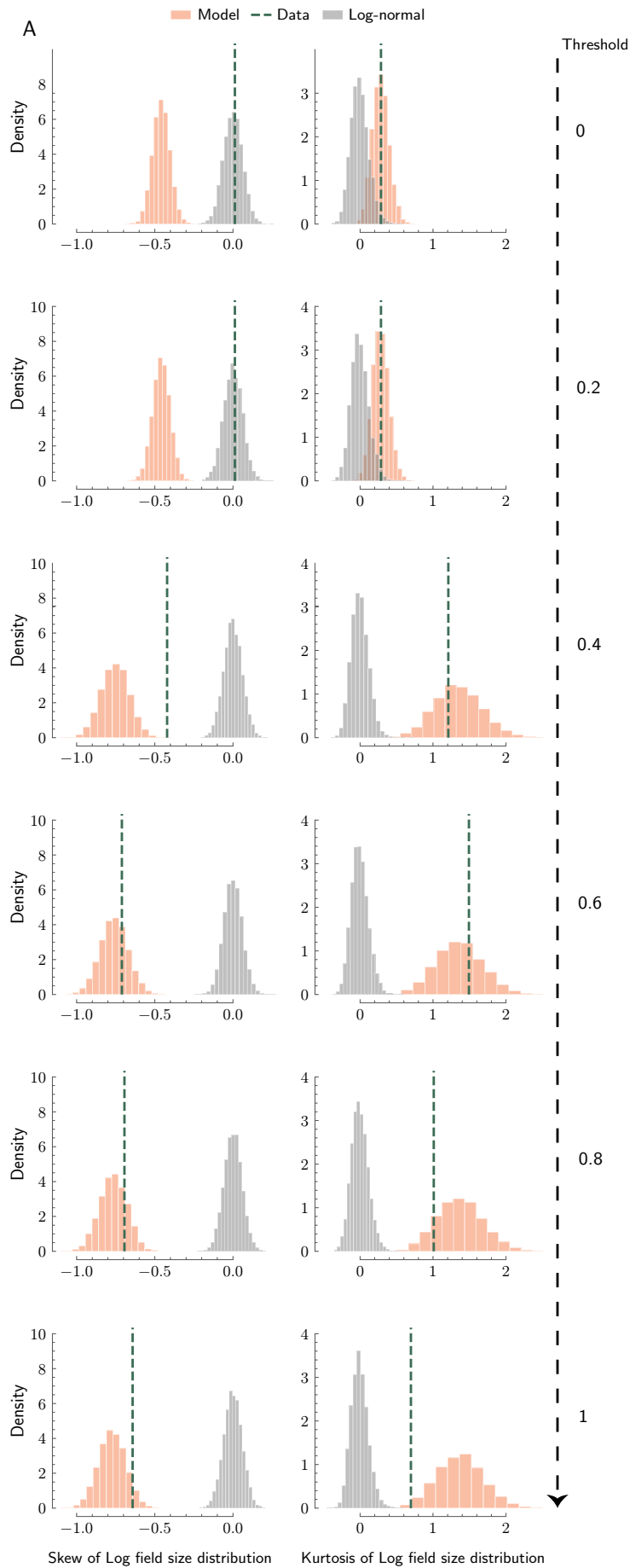


Figure S5. Evolution of field size distribution higher moments and inferred field size-amplitude slope across thresholds. Related to STAR Methods.

(A) The Higher moment test of skew (left column) and kurtosis (right column) between the model and log-normal is highly sensitive and changes across choices of a range of small (between 0 – 1 Hz) thresholds applied to the data for denoising. The choice of denoising threshold for the main figures was 0.5, and the figure shows the tests for 6 other choices.

(B) Slope of the linear fit between log field size and log maximum firing rate, evaluated for the data from bats flying in the 200 m tunnel using a range of rectification parameters. For rectification parameters above 1 Hz the slope rapidly stabilizes on a consistent value.

Methods S1

Introduction to Gaussian process. Related to Figure 1.

In this section we provide a basic introduction to Gaussian Process, and discuss the mapping between the thresholded Gaussian process and a simplified picture of neural dynamics.

Gaussian processes can be viewed as a generalization of a multivariate Gaussian variable, having a probability density function of the form

$$\phi(\mathbf{f}) = \frac{1}{(2\pi)^{L/2} |C|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{f} - \mathbf{m})^\top C^{-1} (\mathbf{f} - \mathbf{m}) \right), \quad (\text{S1})$$

where $\mathbf{f}$ is a multidimensional random variable of dimension L , $\mathbf{m}$ is its mean, C is the covariance matrix and $|C| = \det C$ is the determinant of the matrix C . Multivariate Gaussian variables have the property that for every $\alpha = (\alpha_1, \dots, \alpha_L) \in \mathbb{R}^L$, the real valued variable $\alpha \cdot \mathbf{f} = \sum_{i=1}^L \alpha_i f_i$ is distributed as a univariate Gaussian variable. This property is sufficient and necessary for a multivariate variable to have the form of Eq. S10, and can therefore be used as a definition of a multivariate Gaussian distribution, instead of Eq. S10.

It is possible to generalize the concept of a Gaussian distribution over vectors to a Gaussian distribution over *functions* of a continuous variable $x \in [0, L]$: the variable x can be thought of as a continuous analogue of the discrete index that labels the elements of a random vector. More formally a random distribution over functions $f(x)$ is a Gaussian process if for every discrete set of k points, the values of the function $(f(x_1), \dots, f(x_K))$ are jointly distributed as a multivariate Gaussian. Similarly to the discrete case, where the vector $\mathbf{m}$ and the covariance matrix C uniquely determine the distribution of a multivariate Gaussian variable, a Gaussian process is uniquely determined by its mean function

$$m(x) = \langle f(x) \rangle,$$

and its (two-variable) covariance function

$$r(x, y) = \langle [f(x) - m(x)] [f(y) - m(y)] \rangle.$$

In this work we consider stationary Gaussian processes, having the property that $r(x, y)$ is function of $x - y$ and can thus be written as $r(x - y)$. In addition, the mean $m(x)$ of a stationary Gaussian process is independent of x , and is assumed in this work to vanish. Note, specifically, that the variance of the process is constant throughout the space and is given by $r(0)$.

In our model, the underlying index $x \in [0, L]$ represents the location of the animal in its environment, and the turning curve of a neuron is obtained by applying a threshold nonlinearity to its associated instance of the Gaussian process. A simple biological interpretation of the model is that the Gaussian process represents the mean synaptic input that the neuron receives from its presynaptic partners, whose activities are themselves spatially tuned. The threshold-linear transfer function, which relates the total synaptic input to the cell's firing rate arises from the interplay of the synaptic input, the intrinsic dynamics of the neuron, and noise^{S6}. Technically, in the theory of Gaussian processes, the regions in which the process exceeds a threshold are called excursion sets, and these are mapped to place fields in our model.

According to the model, at any fixed position of the animal the synaptic activation is distributed as a Gaussian variable across the neural population, and so is the synaptic activation of a single neuron across positions. This gives rise to a thresholded Gaussian distribution of firing rates. Interestingly, it was noted long ago in another brain area (IT cortex) that the distribution of firing rates is well approximated by a truncated Gaussian distribution^{S7}.

Methods S2

Derivation of Gaussian process input to CA1 cells. Related to Figure 1.

Here we illustrate how random summation of presynaptic inputs into CA1 cells leads to Gaussian statistics of the total synaptic input, and elucidate how the spatial correlation function of the Gaussian process is determined by the input statistics. We assume, for simplicity, completely independent weights, yet Gaussian statistics can arise also under broader conditions, in which there is partial correlation between groups of summed variables that decays sufficiently rapidly across a functional axis within the input population^{S8}.

Consider a population $i = 1 \dots N$ of spatially selective neurons presynaptic to a CA1 place cell with bounded spatial response function $u_i(x)$ for $x \in \mathbb{R}^D$ in D dimensions. For simplicity we consider these response functions to have zero mean, whereas in a more realistic setting u_i can stand for the deviation of the firing rate from its mean, and approximate balancing of the mean (which is interchangeable also with the choice of threshold in our model) may be obtained by other mechanisms, such as mutual lateral inhibition or homeostatic regulation of excitatory and inhibitory inputs.

We consider first, for simplicity, a scenario in which all input neurons have the same tuning curve up to translation, with preferred firing locations x_i that uniformly and densely tile the space:

$$u_i(x) = u_0(x - x_i). \quad (\text{S2})$$

The input to the CA1 cell, $h(x)$ is a weighted linear sum of the activity of the presynaptic cell outputs

$$h(x) = \sum_{i=1}^N W_i u_i(x), \quad (\text{S3})$$

where we assume that the weights W_i are drawn independently from a random distribution $p(W)$ with zero mean, and a finite second moment. Using the central limit theorem for multidimensional stimuli^{S9,10}, it is straightforward to see that for any discrete set of locations x_α , $h(x_i)$ are jointly distributed as multivariate normal in the large N limit. Hence, $h(x)$ is a Gaussian process in this limit.

The covariance function of the Gaussian process (which uniquely determines all of its properties) must be translationally invariant, due to our assumption of uniform tiling of inputs in the input layer. Indeed,

$$\begin{aligned} \langle h(x)h(x') \rangle &= \sum_{i,j=1}^N \langle W_i W_j \rangle u_i(x) u_j(x') = \text{var}(w) \sum_{i=1}^N u_i(x) u_i(x') \\ &= \text{var}(w) \cdot r(x' - x), \end{aligned} \quad (\text{S4})$$

where the angular brackets represent an average over realizations of the random synaptic weights, and r is the spatial covariance of the neural population response. Here, r is proportional to the auto-correlation of the single neuron tuning curve

$$r(\Delta x) = \rho \int dx u_0(x) u_0(x + \Delta x), \quad (\text{S5})$$

where ρ is the spatial density of receptive field centers. Specifically, if individual tuning curves are shaped as Gaussian firing fields, $r(\Delta x)$ is a Gaussian with a width equal to twice the width of the individual firing fields.

Heterogeneous tuning curves

Similar results hold for heterogeneous tuning curves. We assume that the tuning curves of individual neurons are sampled from a distribution of shapes, parameterized by a random variable α which is independently chosen for each neuron (α can be scalar or multidimensional, discrete or continuous). For each neuron i ,

$$u_i(x) = u_0(x - x_i; \alpha_i), \quad (\text{S6})$$

where x_i is drawn from a uniform distribution in space and α_i is the shape parameter of neuron i . Under these assumptions, $h(x)$ is a sum over many independent random functions, which becomes distributed in the large N limit as a Gaussian process due to the central limit theory. A similar derivation as above yields

$$\langle h(x)h(x') \rangle = \text{var}(w) \cdot r(x' - x), \quad (\text{S7})$$

where $r(\Delta x)$ is proportional to the auto-correlation of the tuning curve, averaged over the distribution of receptive field shapes

$$r(\Delta x) = \rho \left\langle \int dx u_0(x; \alpha) u_0(x + \Delta x; \alpha) \right\rangle_{\alpha}, \quad (\text{S8})$$

where $\langle \rangle_{\alpha}$ represents an average over realizations of the shape parameter α . Qualitatively, in both scenarios discussed above, the central limit theory becomes valid when each CA1 neuron receives, within its place fields, active presynaptic inputs from many spatially selective cells.

Methods S3

Field statistics. Related to Figures 1 and 2.

An extensive body of mathematical results exist for the statistics of excursion sets of Gaussian processes. In this section we explain how a small subset of the results are obtained. Full derivations of all the results are found in Adler^{S11}.

Field statistics in 1D

Field counts in 1d can be exactly derived using the Kac-Rice formula^{S12}:

$$\langle N \rangle = \frac{1}{2} \int_L \left\langle |f'(x)| \mid f(x) = \theta \sqrt{r(0)} \right\rangle dx, \quad (\text{S9})$$

where $f(x)$ is a random process with a well defined derivative. Specifically, in our model $f(x)$ is a 0 mean, stationary Gaussian process, and consequently $f'(x)$ is a 0 mean stationary Gaussian process as well. To calculate the integrand, we first compute the covariance matrix associated with the joint distribution of $f(x)$ and $f'(x)$:

$$C_x = \begin{pmatrix} \langle f(x)^2 \rangle & \langle f(x)f'(x) \rangle \\ \langle f'(x)f(x) \rangle & \langle f'(x)^2 \rangle \end{pmatrix}. \quad (\text{S10})$$

Using stationarity, we have:

$$(i) \quad \langle f(x)^2 \rangle = r(0)$$

$$(ii) \quad \langle f(x)f'(x) \rangle = \frac{1}{2} \frac{\partial^2}{\partial x^2} \langle f(x)^2 \rangle = \frac{1}{2} \frac{\partial^2}{\partial x^2} r(0) = 0$$

$$(iii) \quad \langle f'(x)^2 \rangle = \frac{\partial^2}{\partial x_1 \partial x_2} r(x_1 - x_2) \Big|_{x_1=x_2} = -r''(0).$$

Thus:

$$C_x = \begin{pmatrix} r(0) & 0 \\ 0 & -r''(0) \end{pmatrix}. \quad (\text{S11})$$

Using Eq. S20, we can calculate the expectation inside the integral

$$\begin{aligned} & \langle |f'(x)| \mid f(x) = \theta\sqrt{r(0)} \rangle \\ &= \frac{1}{2\pi\sqrt{-r(0)r''(0)}} \int_{-\infty}^{\infty} |f'(x)| \exp\left(-\frac{-r''(0)f(x)^2 + r(0)f'(x)^2}{-2r''(0)r(0)}\right) \delta(f(x) = \theta\sqrt{r(0)}) df df' \\ &= \frac{e^{-\theta^2/2}}{\sqrt{2\pi r(0)}} \cdot \frac{1}{\sqrt{2\pi r''(0)}} \int_{-\infty}^{\infty} |f'(x)| \exp\left(-\frac{1}{-2r''(0)} f'(x)^2\right) df' \\ &= \frac{e^{-\theta^2/2}}{\sqrt{2\pi r(0)}} \cdot \sqrt{\frac{-2r''(0)}{\pi}} \\ &= \frac{e^{-\theta^2/2}}{\pi} \sqrt{\frac{-r''(0)}{r(0)}}. \end{aligned}$$

Therefore, the final result for $\langle N \rangle$ is

$$\langle N \rangle = \frac{1}{2\pi} \int_L \sqrt{\frac{-r''(0)}{r(0)}} e^{-\theta^2/2} dx = \frac{Le^{-\theta^2/2}}{2\pi\sigma}. \quad (\text{S12})$$

We can directly relate this to the measure of propensity (number of place fields per unit length):

$$\langle \text{Propensity} \rangle = \frac{e^{-\theta^2/2}}{2\pi\sigma}. \quad (\text{S13})$$

Thus, the average propensity is suppressed by both the correlation length σ (algebraically) and the normalized threshold θ (exponentially)

Next, assuming that the environment is large compared to the correlation length, the portion of the environment in which a cell is active is $L(1 - \phi(\theta))$. This, together with the mean field count can be used to calculate the mean field size s and the mean gap between the fields $\bar{s}$ as

$$\langle s \rangle = 2\pi\sigma(1 - \phi(\theta))e^{\frac{\theta^2}{2}}, \quad (\text{S14})$$

$$\langle \bar{s} \rangle = 2\pi\sigma\phi(\theta)e^{\frac{\theta^2}{2}}. \quad (\text{S15})$$

Simulations were in excellent agreement with all these expressions.

Field statistics in higher dimensions

In higher dimensions, precise analytical expressions are not available for the mean field count and size. We relied on simulations to determine the parameters σ and θ that match the mean size and the mean field count, as described in Methods. A qualitative understanding of these relationships can be obtained using an approximation which is valid for high thresholds, obtained

by retaining the leading order term in the expression for the Euler characteristics (see ^{S11} for more details on this approximation):

$$\langle N \rangle = \frac{(L/\sigma)^D}{(2\pi)^{\frac{D+1}{2}}} \theta^{D-1} \exp\left(-\frac{\theta^2}{2}\right), \quad (\text{S16})$$

where L^D is the volume of the environment.

The mean field size is then approximated by

$$\langle s \rangle = \sqrt{2\pi} (2\pi\sigma^2)^{\frac{D}{2}} \theta^{1-D} (1 - \phi(\theta)) e^{\frac{\theta^2}{2}}. \quad (\text{S17})$$

Distributions in the high threshold regime

As described in the main text, distributions related to the field arrangements acquire a universal form in the high threshold regime. The field count is Poisson, in all dimensions, with a mean given by Eq. S23. This is equivalent to the fact that the field gaps in 1d are exponentially distributed. Similarly we can approximate the full field-size distribution as in the main text (Eq. 1) by:

$$P(s) = \frac{2\beta}{D} s^{\left(\frac{2}{D}-1\right)} \exp\left(-\beta s^{\frac{2}{D}}\right).$$

Furthermore, we can relate β to the parameters σ and θ of the model through the analytic forms for the mean field size (Eqs. S22 and S24). To that end, first, observe that $s^{2/D}$ is exponentially distributed with an expectation $\langle s^{2/D} \rangle = 1/\beta$, implying that $\langle s \rangle = \Gamma(D/2 + 1)\beta^{-D/2}$ using the moment generating function of the exponential distribution. Therefore:

$$\beta = \left[\Gamma\left(\frac{D}{2} + 1\right) \langle s \rangle \right]^{\frac{2}{D}}. \quad (\text{S18})$$

Methods S4

Convergence to Rayleigh distribution. Related to Figures 1, 2 and 5.

The field size distribution in our model has a stereotypical skewed form that converges to a Rayleigh distribution in the high threshold limit. To understand this behavior on very qualitative terms, we note that for high threshold crossings the fields are small compared to the correlation length of the Gaussian process, and the consecutive threshold crossings demarcating the fields are correlated (contrast this with field gaps, which in the high threshold limit are large compared to the correlation length of the process. The consecutive threshold crossings demarcating gaps are therefore *uncorrelated* in the high threshold limit, thereby leading to the observed exponential distribution).

A full mathematical characterization of the threshold crossing's joint distribution – when the crossings are nearby as in the case of the place fields, is still an open problem for general Gaussian processes, yet some results are available. Under broad regularity conditions, it has been mathematically established that, when the threshold crossings are nearby, the threshold crossings tend to repel each other ^{S13}. Moreover, it is established that the probability of small distance δx between the consecutive nearby threshold crossing $p(\delta x)$ behaves, for small δx as

$$p(\delta x) \propto \delta x. \quad (\text{S19})$$

Effectively this implies that the probability of small field size approaches 0 when $\delta x \rightarrow 0$. The repulsion between consecutive crossings can be understood intuitively from the observation that

for there to be another threshold crossing after a threshold crossing, the derivative of the process must change sign. Since we assume a smooth process, the sign change of the derivative will only occur after some distance from the first threshold crossing, effectively suppressing the probability of having very small excursion sets.

It is also straightforward to evaluate how the probability density function for the gap size x decays to zero for large x . Suppose that a stationary Gaussian process $f(x)$ with unit variance crossed the threshold θ at the origin i.e, $f(0) = \theta$. We will calculate the probability that the process is still above the threshold at x , i.e, $P\{f(x) > \theta \mid f(0) = \theta\}$. To that end, we start by noting that $f(0)$ and $f(x)$ are jointly Gaussian with covariance $r(x)$, and by Gaussian conditioning we get:

$$(f(x) \mid f(0) = \theta) \sim \mathcal{N}(\theta r(x), 1 - r(x)^2). \quad (\text{S20})$$

We can then express the probability of $f(x)$ being above threshold by

$$\begin{aligned} P\{f(x) > \theta \mid f(0) = \theta\} &= 1 - \Phi\left(\frac{\theta - \theta r(x)}{\sqrt{1 - r(x)^2}}\right) \\ &\leq \mathcal{O}\left(\exp\left[-\frac{1}{2} \frac{\theta^2 (1 - r(x))^2}{1 - r(x)^2}\right]\right) \\ &= \mathcal{O}\left(\exp\left[-\frac{\theta^2}{2} \frac{1 - r(x)}{1 + r(x)}\right]\right) \\ &\approx \mathcal{O}\left(\exp\left[-\frac{\theta^2}{4} \left(\frac{x}{\sigma}\right)^2\right]\right) \\ \implies P\{f(x) > \theta \mid f(0) = \theta\} &\leq \mathcal{O}\left(\exp\left[-\left(\frac{\theta x}{2\sigma}\right)^2\right]\right), \end{aligned} \quad (\text{S21})$$

where we have used the Taylor expansion of the correlation function $r(x) \approx 1 - \frac{1}{2} \left(\frac{x}{\sigma}\right)^2$ which holds in the high threshold regime. We can conclude from this derivation that the tail of the field size distribution decays as a function of x at least as fast as $\exp\left[-\left(\frac{\theta x}{2\sigma}\right)^2\right]$.

Together, these results on the dependence of the probability density function on x – for small x (linear dependence on x) and for large x (decay as a squared exponential function of x) explain the key features of the Rayleigh distribution. Next, we will sketch a more formal derivation of the field size distribution in the high threshold limit.

We start with a theorem concerning the behavior of a Gaussian process near a maximum that occurs above a high threshold^{S11}. Consider a centered stationary Gaussian process $f(x)$ with unit variance. Assume that $f(x)$ has a local maximum at $x = 0$ that exceeds a threshold θ by a height h , such that $f(0) = \theta + h$. When θ approaches infinity and for small values of $|x|$, the process can be approximated as

$$f(x) \approx f(0) + f(0) \frac{r''(0)}{2} x^2. \quad (\text{S22})$$

This theorem simply states that at the limit of a high threshold crossing, in the neighborhood of the maxima of the firing rate, the fields are well approximated by a parabola with curvature that concentrates at $r''(0)$. This allows us to think of the field as parabolic and express the relationship between the size of the excursion s and the field height h :

$$\lim_{\theta \rightarrow \infty} s = 2\sigma \sqrt{\frac{h}{\theta}}. \quad (\text{S23})$$

(Note that this results implies a power law relating the field size and amplitude with exponent 2, in agreement with the trends seen in Fig. 5. Note also that field shapes can be well approximated by a parabola for $\theta = 2.4$.)

Equipped with Eq. (S32), if we now find the distribution for the height of the fields h , we can then approximate the distribution of the field sizes. To that end, we define the height variable $H_\theta = f(0) - \theta$ and observe that

$$P\{H_\theta \geq h \mid O_\theta \geq 0\} = \frac{\langle M(\theta + h) \rangle}{\langle M(\theta) \rangle}, \quad (\text{S24})$$

where $\langle M(\theta) \rangle$ represents mean number of maxima of a Gaussian process above the threshold θ . Intuitively, the expression relates the tail of the height distribution to the ratio between the mean number of maxima above $\theta + h$ and the mean number of maxima above θ .

In the high threshold regime, each field has a single maximum, and we can replace the mean number of maxima $M(\theta)$ with the field count $N(\theta)$, which we know from the Kac-Rice formula, Eq. (S21). We thus obtain:

$$P\{H_\theta \geq h \mid O_\theta \geq 0\} = \frac{\exp(-(\theta + h)^2/2)}{\exp(-\theta^2/2)},$$

and in the high threshold limit,

$$\begin{aligned} P\{H_\theta \geq h \mid O_\theta \geq 0\} &= \frac{\exp(-(\theta + h)^2/2)}{\exp(-\theta^2/2)} \\ &= \frac{\exp(-(\theta^2 + 2h\theta + h^2)/2)}{\exp(-\theta^2/2)} \\ &= \frac{\exp(-\theta^2/2) \cdot \exp(-(2h + h^2/\theta^2)/2)}{\exp(-\theta^2/2)} \\ &= \exp(-h\theta + h^2/2). \end{aligned}$$

Viewed as a function of $\hat{h} = \theta h$, the distribution approaches a non-trivial stationary limit when $\theta \rightarrow \infty$,

$$\lim_{\theta \rightarrow \infty} P\{H_\theta \geq \hat{h}/\theta \mid O_\theta \geq 0\} = \lim_{\theta \rightarrow \infty} \exp(-\hat{h} + \hat{h}^2/2\theta^2) = \exp(-\hat{h}). \quad (\text{S25})$$

We conclude that the height is distributed as an exponential random variable with mean $1/\theta$, and since the field size is related to the height, we can show using a change of variables to show that the field size is distributed as a Rayleigh distribution. First recall that

$$\lim_{\theta \rightarrow \infty} s = 2\sigma \sqrt{\frac{h}{\theta}} = \frac{2\sigma}{\theta} \sqrt{\hat{h}} \implies \hat{h} = \frac{\theta^2 s^2}{4\sigma^2},$$

where $\sqrt{\hat{h}} \sim \exp(1)$. Then:

$$\begin{aligned} P_{\hat{h}}(\hat{h}) &= e^{-\hat{h}} \quad ; \quad \frac{d\hat{h}}{ds} = \frac{\theta^2 s}{2\sigma^2} \\ \implies P_s(s) &= P_{\hat{h}}(\hat{h}(s)) \left| \frac{d\hat{h}}{ds} \right| = \exp\left(-\frac{\theta^2 s^2}{4\sigma^2}\right) \cdot \frac{\theta^2 s}{2\sigma^2}, \end{aligned}$$

which is a Rayleigh distribution.

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